

Determination of the Half-Beam cone angle for Pierce electron gun design using an artificial neural network

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ABSTRACT

Various electron gun design methods, especially iterative and noniterative, are generally utilized for preliminary electron gun design based on the input design beam parameters. In the electron gun design methods, calculating the half-beam cone angle is crucial, as it significantly influences the determination of the electron gun's geometry. In this paper, an Artificial Neural Network (ANN) model was proposed for predicting the half-beam cone angle. The ANN model was trained and tested using experimental data obtained from previously documented studies on electron guns. The predicted results of the proposed ANN model were compared with three well-known methods in the electron gun design: iterative method, noniterative method, and modified noniterative method. The comparison indicates that the proposed ANN model achieves better performance than the other methods, with a Main Relative Error (MRE) of less than 5%.

Introduction

The electron beam is increasingly used in various fields such as metallurgy [1–4], electron microscopy [5,6], scientific research [7–9], etc. An electron gun, which produces, shapes, and directs the electron beam, is an essential component of any vacuum electronic device, accelerator, synchrotron radiation source, or free electron laser [10–12]. Due to the increasing importance of electron sources in various industrial and research applications, the design and fabrication of the electron gun have gained significant attention [13]. Historically, before the advent of computers, the design of the electron gun was based on the graphical method [14]. Then, the design of the electron gun was carried out using numerical computer calculation with the advancement of computing technology. A preliminary estimation of the electron gun configuration is obtained through the synthesizing method. Nowadays, the initial geometry of the electron gun is typically synthesized using numerical computer calculation. The obtained geometry is used as input for the gun simulation software. According to the simulation result, further refinement takes place on gun geometry to obtain the desired output beam. Refining the electron gun geometry through numerical simulation requires less time and effort if the initial estimation has been made close enough through the synthesis method [15]. In the initial design, most electron guns are designed based on guidelines established

by Pierce and are known as Pierce electron guns [16]. Vaughan presented an iterative method to synthesize the convergent Pierce electron gun with the curved cathode [17]. The inputs of this method are beam voltage (V), current (I), beam waist radius (r_w), and either the operating cathode current density J_c or the beam convergence ratio (C), which is defined here as the ratio of the cathode area to the beam waist area. Also, the current and voltage parameters can be substituted with perveance (P), which relates the current to voltage in the space charge-limited mode and is defined as $P = \frac{I}{V^{3/2}}$. In Vaughan's iterative

method, the half-beam cone angle (θ) is calculated in the iterative procedure. Then with the use of the obtained θ , other geometrical parameters of the electron gun are computed [17]. Sharma et al. [15] improved Vaughan's iterative method for synthesizing the anode aperture. The modified half-beam cone angle and the modified Langmuir-Blodgett function were used to calculate the anode aperture radius. Their findings indicated that this approach better agreed with the actual values compared to several available guns. Furthermore, this method does not require the use of any empirical factors. One of the popular methods for designing electron guns is the non-iterative method, presented by Tiwary and Basu for the design of a convergent Pierce electron gun with a curved cathode [18]. Similar to the iterative method, this approach also utilizes the aforementioned input design parameters and produces the same outputs. In this method, the half-beam cone angle is

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calculated using an analytical method, and without any iterative procedure. The half-beam cone angle is calculated based on a small-angle approximation. The noniterative method demonstrates a relatively satisfactory accuracy when the gun's convergence half angle remains below 20° . Yang et al. [19] improved Tiwary's noniterative method using an empirical factor in the calculation of half-beam cone angle. Several empirical electron guns with various half-beam cone angles and various perviances have been studied for validation checks of their presented method. Their results indicate that Yang's modified noniterative method exhibits higher consistency with the measured values, excluding the anode-cathode spacing. Gharaati and Mardani [11] designed a Pierce electron gun with a tunable cathode to anode distance. This design incorporates a spring-film mechanism that allows the cathode to move ± 1 mm in the z-direction (beam direction), while the other components are fixed and the vacuum of the electron gun chamber does not change. The proposed configuration enables adjustments to critical parameters, including perveance, cathode lifetime, and body current. The advantage of this approach lies in its flexibility; if the measured perveance is not desirable after assembling the electron gun, the desired perveance value can be achieved by refining the cathode to anode distance without disassembling the device. Zhang et al. [20] proposed a design approach for a triode-type Magnetron Injection Gun (MIG) using a genetic algorithm. In their work, an Artificial Neural Network (ANN) and a decision tree were employed to identify the bad parameter sets whose beam transportation rate is poor without the need to run the particle tracing simulation in the optimization process. Their ANN model utilized 15 input parameters, including MIG geometry and magnetic field parameters, while the ANN outputs consisted of bad parameter sets and beam alpha values. By utilizing ANN, their method successfully reduced the number of required particle-tracking simulations by 28 %, significantly decreasing the optimization time. Ou et al. [21] proposed a design and optimization approach for a 0.22-THz TWT electron gun. For optimizing the electron gun, Support Vector Regression (SVR) combined with the goal attainment algorithm is used considering the Periodic Permanent Magnet (PPM). Both simulation and experimental results demonstrate that the transmission rate of the designed electron gun reaches 95 %. The effectiveness of the proposed method was validated through a prototype electron gun and PPM system.

Among the output parameters of electron gun design, the half-beam cone angle is critically important, as it significantly influences the determination of the electron gun's geometry. Therefore, we aimed to determine the half-beam cone angle using Artificial Neural Network (ANN). ANNs can be defined as mathematical systems that mimic the biological nervous system [19]. ANNs are particularly powerful tools for classification and prediction in many models [22–25]. Therefore, ANN can be a suitable candidate to predict half-beam cone angle in the design of an electron gun. The ANNs are employed to generate outputs based on specific training data. One advantage of the ANN approach is that the majority of complex computations are conducted during the training phase. Once the ANN has been trained for a specific task, the operation is fast, and unknown samples can be rapidly estimated in the operational domain [26]. In this work, the ANN is used to predict the half-beam cone angle in the design of an electron gun. The predicted results with ANN are compared with three aforementioned methods including Vaughan's iterative method, Tiwary's noniterative method, and Yang's modified noniterative method.

Methodology

The half-beam cone angle is one of the most important output parameters in the electron gun design methods. The geometry of an electron gun can be determined by the half-beam cone angle and the parameters employed as inputs for gun design. However, in most electron gun design methods, empirical factors are generally used to guide

the design process. Fig. 1 illustrates the schematic subcomponents of the Pierce electron gun and their synthesized parameters. Also, the key relationships governing electron gun configurations, which play an important role in defining the geometry are presented in equations (1) to (7).

$$r_c = \sqrt{\frac{I}{\pi J_c}} \quad (1)$$

$$r_c = \sqrt{C} r_w \quad (2)$$

$$P = \frac{I}{V^{3/2}} \quad (3)$$

$$R_c = \frac{r_c}{\sin(\theta)} \quad (4)$$

$$R_a = R_c e^{-\gamma} \quad (5)$$

$$r_b(z_a) = r_c e^{-\gamma} \quad (6)$$

$$z_{ac} = R_c - R_a \quad (7)$$

where r_c is the cathode disk radius, R_c is the cathode spherical radius, J_c is cathode current density, P is the perveance, C is the convergence, r_w is the beam waist radius, $r_b(z_a)$ is beam radius at anode aperture, R_a anode spherical radius, z_{ac} is cathode to anode distance, and $\gamma = Ln \frac{R_c}{R_a}$. Determining the half-beam cone angle is the most important in the electron gun design methods. Since the geometry of the electron gun can be determined using the half-beam cone angle. In this work, we focused on the prediction of half-beam cone angle using ANN.

ANN

ANNs are mathematical models composed of simple processing units named neurons that operate in parallel [27]. Each ANN typically includes three main parts: the connection pattern among the neurons, the method for calculating the weights of these connections, and the activation function.

During the training process of the ANN, the connection weights are iteratively adjusted to minimize the prediction error and enhance the model's accuracy. Additionally, bias terms are incorporated into each neuron to provide additional flexibility, enabling the network to better capture nonlinear relationships in the data. Among the different types of ANNs, Multi-layer Perceptron (MLP) networks are the most extensively applied [28]. MLP is a feed-forward neural network, consisting of an input layer, one or more hidden layers, and an output layer [29]. Feature scaling is a vital step in preprocessing data before building a model using ANN [30]. It is especially required when features have a different range of values. It involves adjusting the scale of feature values to facilitate easier processing [31]. Normalization is a common method of feature scaling, where the values are typically mapped between -1 and 1 [32,33]. In this work, all datasets are normalized to the range of $[-1, 1]$. In the present work, the half-beam cone angle values were predicted using the MLP model. In this study, several ANN architectures were examined to identify the most effective configuration. The input parameters selected for the model included perveance, beam waist radius, and convergence. Different architectures, consisting of one, two, and three hidden layers with varying numbers of neurons in each layer, were tested. Among the configurations explored, the one illustrated in Fig. 2 was selected for this work. Although we do not assert that this configuration is the optimal choice, it has achieved reliable results. Table 1 provides the details of the proposed ANN model used in this study. The proposed MLP model involves two hidden layers with 3 neurons in the first hidden layer and 2 neurons in the second hidden layer.

The proposed MLP network was trained utilizing the Levenberg-

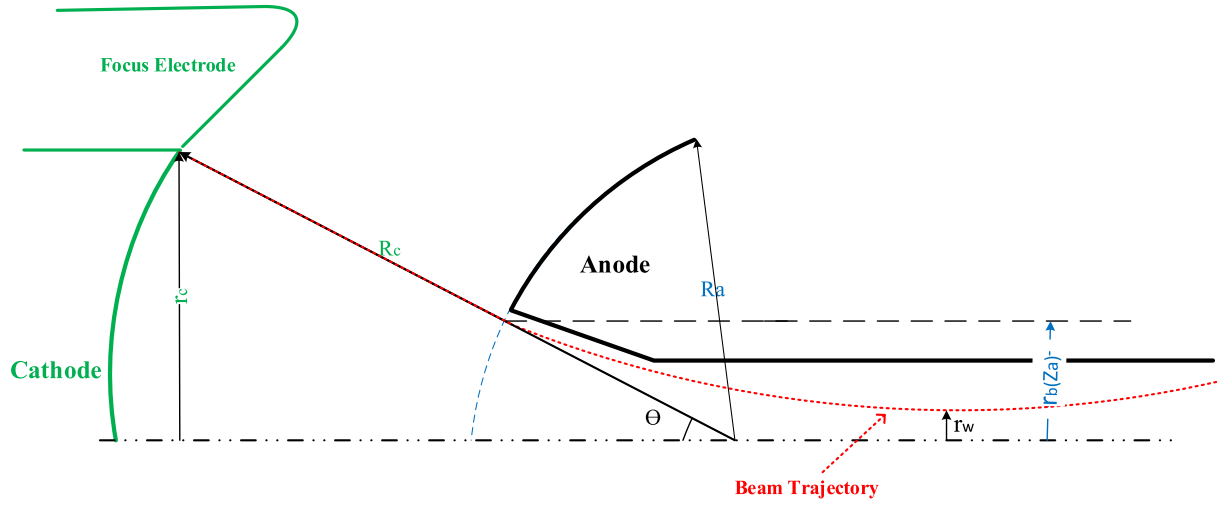


Fig. 1. Geometrical view of Pierce electron gun.

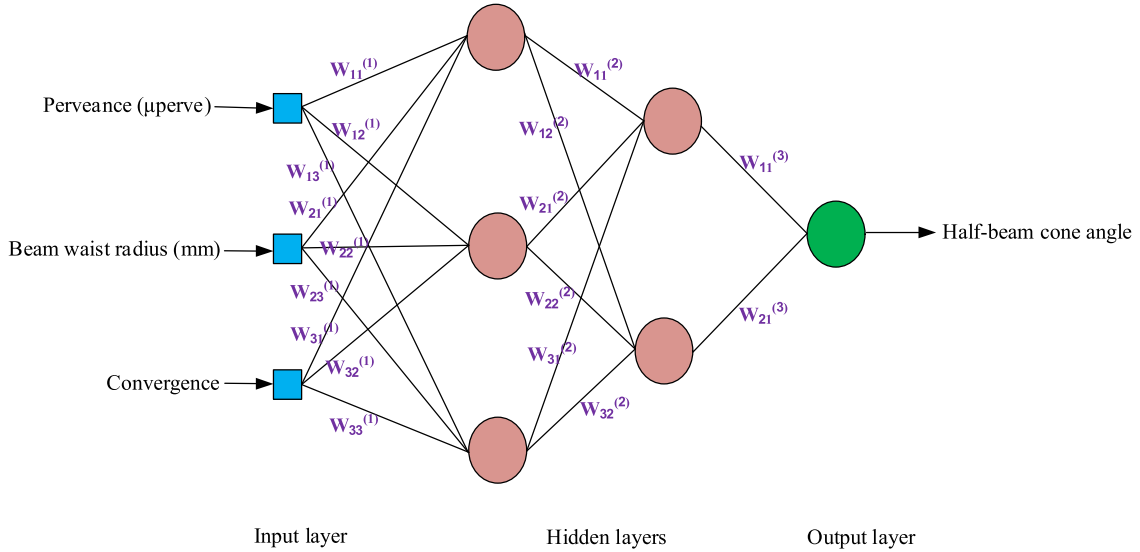


Fig. 2. The architecture of the proposed MLP model.

Table 1
Detail of proposed ANN model.

Neural network	MLP
Number of neurons in the input layer	3
Number of neurons in the first hidden layer	3
Number of neurons in the second hidden layer	2
Number of neurons in the output layer	1
Number of epochs	5000
Activation function	tansig

Marquardt (LM) algorithm. The LM algorithm incorporates two numerical minimization techniques: the gradient descent method and the Gauss-Newton method. In the gradient descent, the sum of the squared errors is reduced by updating the coefficients along the direction of the steepest descent. The Gauss-Newton method minimizes the sum of squared errors by locally approximating the least squares function as quadratic, and then identifying the minimum of this quadratic approximation. The Levenberg-Marquardt method behaves like a gradient descent method when the coefficients are far from their optimal values and transitions to the behavior of the Gauss-Newton method as the coefficients approach their optimal values [34]. The overall performance

of the MLP and other electron gun design methods can be evaluated using mean relative error (MRE), mean square error (MSE), and root mean square error (RMSE) as defined in equations (8) to (10), respectively [27,35].

$$\text{MRE}(\%) = 100 \times \frac{1}{N} \sum_{i=1}^N \left| \frac{\theta_i(\text{Exp}) - \theta_i(\text{Pred})}{\theta_i(\text{Exp})} \right| \quad (8)$$

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (\theta_i(\text{Exp}) - \theta_i(\text{Pred}))^2 \quad (9)$$

$$\text{RMSE} = \left[\frac{\sum_{i=1}^N (\theta_i(\text{Exp}) - \theta_i(\text{Pred}))^2}{N} \right]^{0.5} \quad (10)$$

where N is the number of data points, $\theta(\text{Exp})$ is the desired target which is given by experimental data, and $\theta(\text{Pred})$ refers to predicted or calculated values. The proposed ANN model was trained and tested using 30 data points. These data were selected from experimental guns referenced in the literature and are presented in Table 2. Specifically, cases 1, 2, 9, 12, 25, and 29 are extracted from references [14,18,19], whereas the remaining cases are obtained from references [19]. The

Table 2

Comparison between experimental and calculated results for various methods.

Case number	Perveance (μ perv)	r_w (mm)	C	θ Experimental (degree)	θ predicted by ANN (degree)	θ Iterative method** (degree)	θ Noniterative method (degree)	θ modified Noniterative method (degree)
1	3.00	0.62	207.60	70.00	69.99	69.21	45.03	66.60
2	2.20	0.51	306.30	57.30	57.30	57.26	39.16	57.43
3*	2.50	2.49	16.13	50.28	45.03	41.40	30.75	44.68
4	3.67	4.33	5.33	43.98	43.98	39.04	29.37	42.63
5	2.29	2.35	18.11	43.98	43.79	40.27	29.99	43.55
6*	2.27	2.72	13.52	42.16	42.45	37.85	28.36	41.12
7	2.93	4.26	5.51	41.81	39.46	34.86	26.45	38.30
8	2.33	3.23	9.59	39.87	39.76	35.63	26.91	38.98
9	1.90	1.35	35.55	38.90	39.43	40.46	30.35	44.09
10	3.27	4.42	5.12	38.12	40.40	36.31	27.34	39.62
11	1.32	1.41	50.30	37.04	36.10	34.60	26.41	38.24
12*	1.20	0.76	111.00	35.70	36.09	35.99	27.13	39.30
13	1.52	2.81	12.66	35.08	33.52	30.14	22.85	33.00
14	1.34	1.95	26.30	33.96	35.03	31.92	24.27	35.08
15*	1.01	2.00	25.00	30.00	29.74	27.25	20.86	30.09
16	1.32	2.93	11.65	30.00	31.30	27.49	20.95	30.22
17*	0.74	1.23	66.10	29.04	27.83	26.21	20.32	29.29
18	1.29	4.05	6.10	26.01	26.01	23.28	17.97	25.88
19	1.07	3.71	7.27	26.39	26.39	22.12	17.09	24.60
20	0.79	2.03	24.27	26.01	25.86	23.86	18.34	26.42
21	0.77	1.83	29.86	24.96	26.21	24.26	18.70	26.94
22	0.53	1.82	30.19	22.99	21.76	20.03	15.52	22.32
23	0.32	1.46	46.91	18.45	18.73	16.36	12.80	18.40
24	0.30	2.01	24.75	17.42	17.55	14.57	11.31	16.24
25	0.55	0.94	11.41	17.40	16.66	17.42	13.43	19.30
26*	0.29	1.46	46.91	17.00	18.12	15.56	12.19	17.51
27	0.79	4.13	5.86	16.41	16.41	17.95	13.89	19.97
28	0.33	4.23	5.59	13.02	13.02	11.40	8.84	12.69
29*	0.40	0.84	10.61	12.70	12.40	14.59	11.29	16.21
30	0.38	0.77	9.31	9.80	10.26	13.84	10.72	15.40

Note: * Test data. ** True modification has been applied ($\frac{\sin(\theta)}{\sin(\theta_T)} = 0.905$).

half-beam cone angles of the selected practical guns for ANN training and testing range from 9.80° to 70.00° . Furthermore, the perveance of the selected guns varies from 0.3 to $3.67 \mu\text{P}$.

Results and discussion

The main goal of this work is to estimate the half-beam cone angle of the electron gun. The number of data used for training was 23 (about 76 % of the data set) whereas those used for testing were 7 (about 24 % of the data set). Fig. 3 illustrates the correlation between the experimental data and the predicted results generated by the proposed ANN model for both training and testing datasets in the regression diagrams. The regression diagram is commonly used to show the precision of ANN graphically so that for a more precise network, the data should be as close as possible to the line $y = x$. As depicted in this figure, the predicted values are close to the real values. The numerical comparison of experimental data and predicted results by the proposed ANN model can be found in Table 2. The Pearson correlation coefficient is a dimensionless index that measures the strength and direction of a linear relation between two variables. The range of the Pearson correlation coefficient is between -1 to $+1$. If the variables are directly related, the sign of correlation coefficient is positive and if the variables are inversely related, the sign is negative. The closer the value of the correlation coefficient to 1, the stronger the measure of closeness to a linear relationship. The Pearson correlation coefficient formula is mentioned in equation (11) [36,37].

$$R = \frac{\sum_{i=1}^n (x_i - \bar{x}) \times \sum_{i=1}^n (y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \times \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \quad (11)$$

where n is the size of the data, x_i and y_i are the individual data points,

and $\bar{x}$ and $\bar{y}$ are the all data averages. According to Eq. (1), the correlation coefficient between experimental data and predicted results by ANN are 0.997 and 0.990 for training and testing data, respectively.

To illustrate the training process of the artificial neural network, the MSE is plotted against the number of epochs in Fig. 4. The training error decreases significantly in the initial stages and gradually stabilizes as the model converges. The training time of the neural network on traditional standard computers is less than a few seconds.

The error values for the best-fit model of the proposed ANN model have been summarized in Table 3. According to this table, the proposed ANN model is a useful and reliable method to predict the half-beam cone angle in the electron gun.

Fig. 5 shows the comparison among the results of the presented ANN model, Vaughan's iterative method, Tiwary's noniterative method, Yang's modified noniterative method, and the associated actual values. Fig. 6 illustrates the error of each method, calculated as the difference between the estimated values and the corresponding actual values. As depicted in these two figures, the predicted half-beam cone angle by the proposed ANN model is closer to the corresponding actual values compared to the results obtained from the other methods. Also, the numerical comparison of the estimated half-beam cone angle by the aforementioned methods is presented in Table II.

The MRE and RMSE values for the test data of all methods, including the proposed ANN model, Vaughan's iterative method, Tiwary's noniterative method, and Yang's modified noniterative method are presented in Table 4. The comparison of MRE and RMSE between the proposed ANN model and other methods demonstrates that the ANN method is a reliable and useful method to predict the half-beam cone angle of the electron gun. The execution time of all examined methods on traditional standard computers is typically less than a few milliseconds, with the non-iterative and modified non-iterative methods being the fastest among them. However, due to better performance of ANN

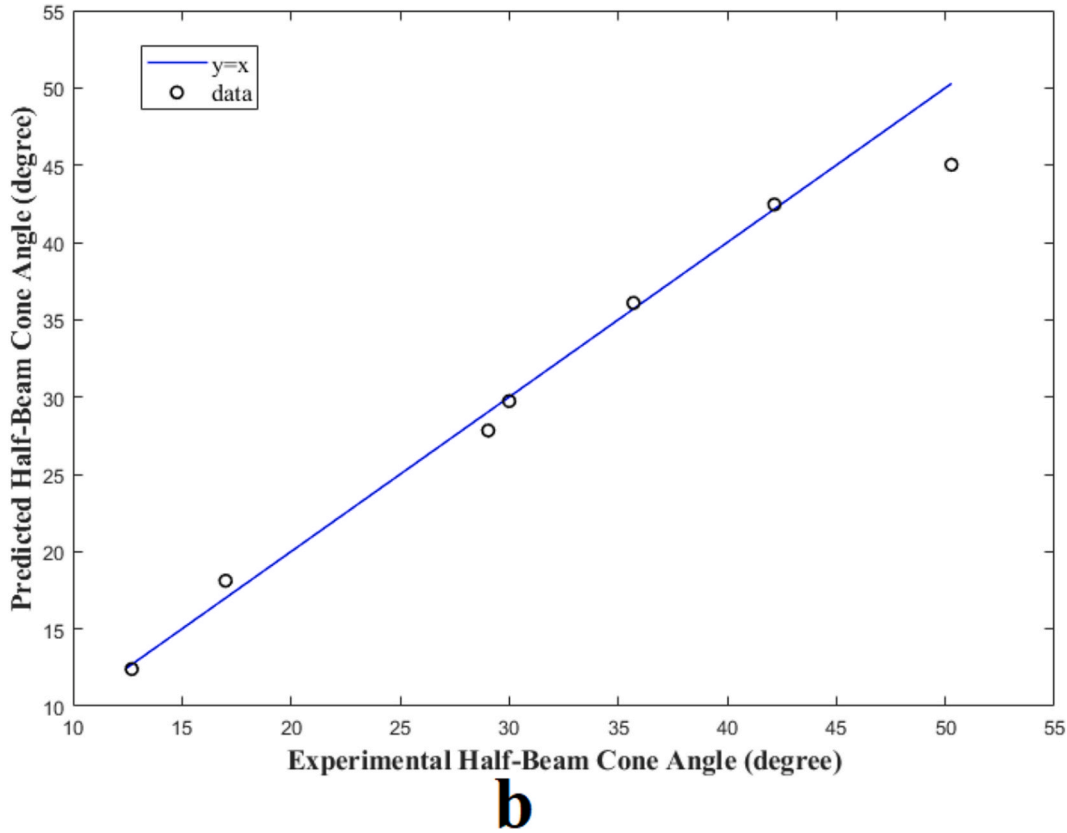
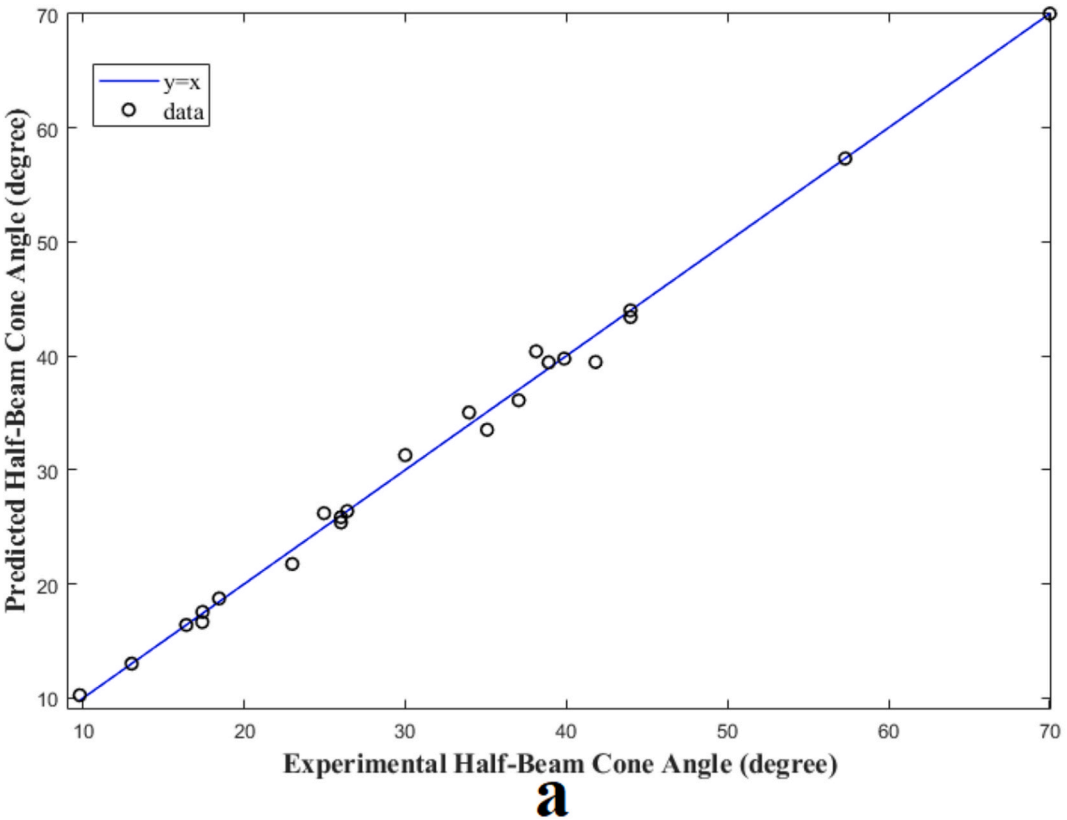


Fig. 3. Regression diagram of experimental and ANN predicted results across different half-beam cone angles: a) training data and b) testing data.

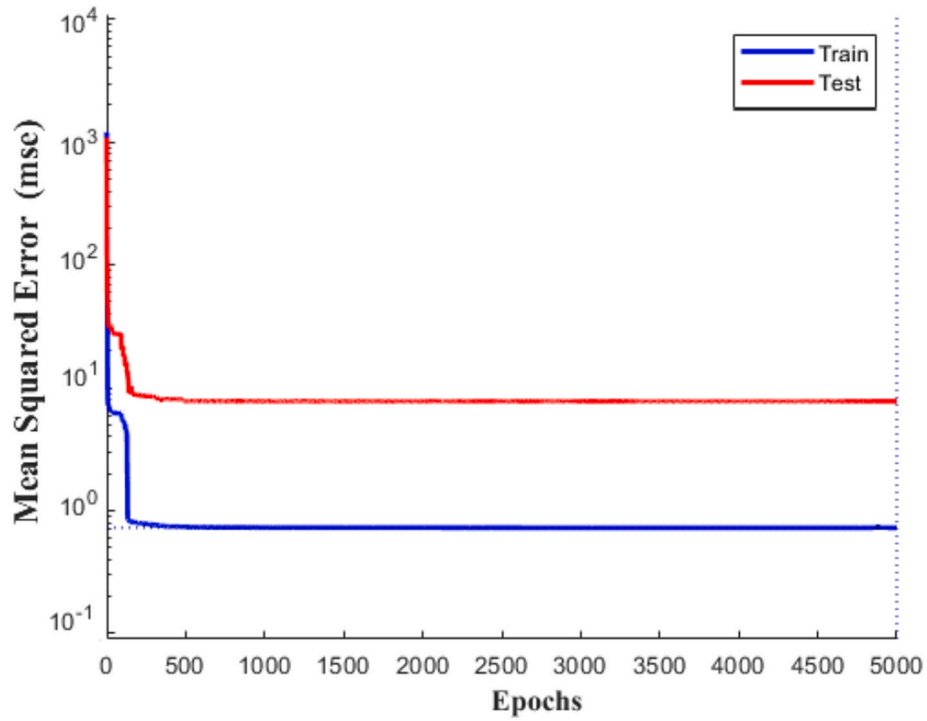


Fig. 4. MSE versus the number of epochs.

Table 3

Obtained errors for training and testing results.

Error	Train	Test
MRE (%)	2.18	3.74
RMSE	0.95	2.09

compared with other method, the ANN is a more suitable tool for electron gun design.

Our results indicate that the artificial neural network can effectively estimate the half-beam cone angle. Other geometric parameters of the electron gun can be determined using the half-beam cone angle and the input parameters of proposed ANN model, as outlined in Equations (1) to (7). Additional relationships can be found in references such as [17].

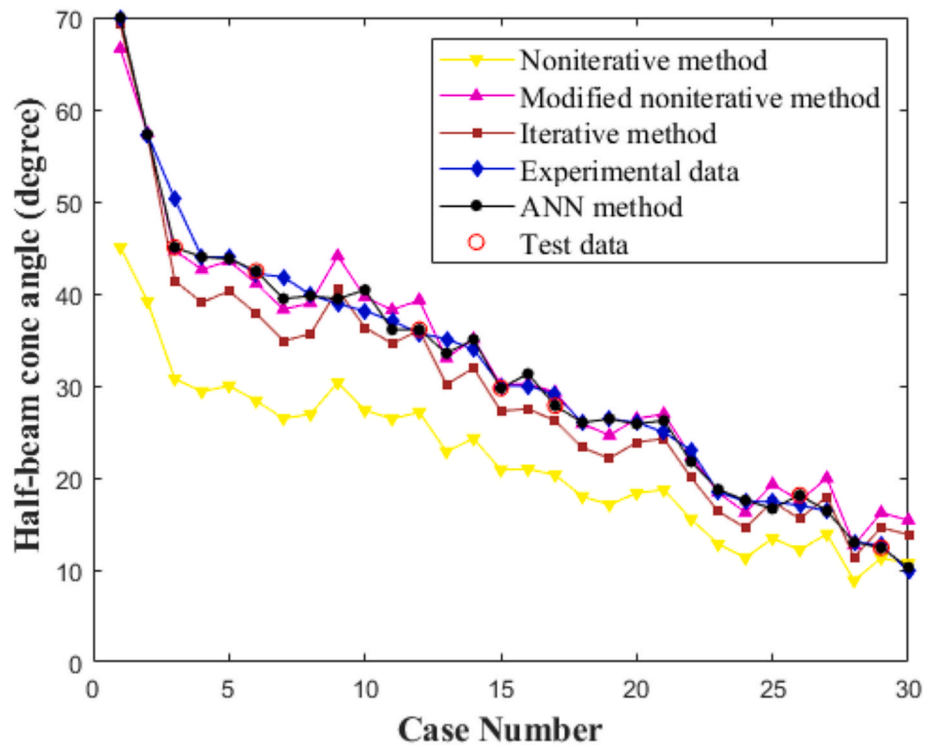


Fig. 5. Comparison of half-beam cone angles calculated by various methods.

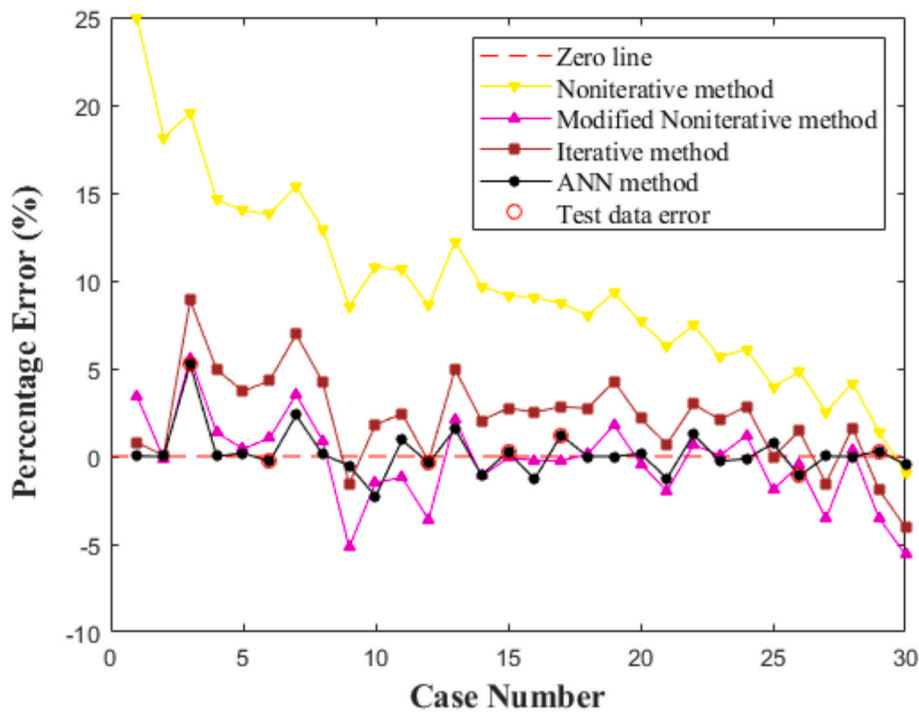


Fig. 6. Percentage error in half-beam cone angle calculations for various methods.

Table 4
Comparison of obtained error for various methods based on the test data.

Method Error	ANN	Iterative method	Noniterative method	Modified noniterative method
MRE (%)	3.74	10.14	27.93	7.93
RMSE	2.09	4.11	10.89	2.88

Thus, the ANN can be used as an effective tool for preliminary design of Pierce electron gun.

Conclusion

The half-beam cone angle plays a vital role in determining the electron gun geometry. In this study, a novel method based on an artificial neural network has been introduced to predict the half-beam cone angle for the initial design of electron guns. The neural network was trained using data from practical electron guns, encompassing an extensive variety of perviance, convergence, and half-beam cone angles, with the half-beam cone angles specifically varying from 9.80° to 70.00°. The proposed ANN model predicted the half-beam cone angle with a mean relative error of less than 5 %. The proposed ANN model was compared with three well-known methods for the design of an electron gun: Vaughan’s iterative method, Tiwary’s noniterative method, and Yang’s modified noniterative method. The comparison reveals that the proposed ANN model has a better performance than the other methods. Consequently, the ANN is a suitable and accurate tool for predicting the half-beam cone angle in the preliminary design of electron guns.

CRedit authorship contribution statement

R. Panahi: Writing – original draft, Methodology, Investigation, Conceptualization. **S.A.H. Fegghi:** Supervision, Formal analysis. **Sh. Sanaye Hajari:** Writing – review & editing, Methodology, Formal analysis. **M. Khorsandi:** Writing – review & editing, Formal analysis,

Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

[1] Iqbal M, et al. Surface modification of mild steel with boron carbide reinforcement by electron beam melting. *Vacuum* 2010;85(1):45–7.

[2] Hasan AM, et al. Optoelectronic properties of electron beam-deposited NiOx thin films for solar cell application. *Results Phys* 2020;17:103122.

[3] Hoseinzade M, Nijat M, Sadighzadeh A. Numerical simulation and design of a thermionic electron gun. *Chin Phys C* 2016;40(5):057003.

[4] Iqbal M, Rafiq M, Bhatti SA. The electron beam gun with thermionic hairpin-like cathode for welding and surface modifications. *Vacuum* 2004;77(1):19–26.

[5] Krysztof M. Field-emission electron gun for a MEMS electron microscope. *Microsyst Nanoeng* 2021;7(1):43.

[6] Song Y, et al. Development of a 1.4-cell RF photocathode gun for single-shot MeV ultrafast electron diffraction devices with femtosecond resolution. *Nucl Instrum Methods Phys Res, Sect A* 2022;1031:166602.

[7] Markov A, Yakovlev E, Bestetti M. Synthesis of a Cr-Cu surface alloy using a low-energy high-current electron beam. *Results Phys* 2019;12:1915–24.

[8] Huang WR, et al. Toward a terahertz-driven electron gun. *Sci Rep* 2015;5(1):14899.

[9] Ploskih AE, et al. Performance improvement of a sub-THz traveling-wave tube by using an electron optic system with a converging sheet electron beam. *Results Phys* 2019;12:799–803.

[10] Iqbal M, et al. Design, simulation and analysis of beam optics and solenoid of high-power gun for RF power source. *Nucl Instrum Methods Phys Res, Sect A* 2021;1014:165703.

[11] Gharaati A, Mardani D. Electrostatic design and fabrication a new tunable perviance pierce electron gun. *IEEE Trans Electron Devices* 2020;68(1):318–23.

[12] Huang P-W, et al. An ultrahigh-vacuum S-band photocathode radio-frequency electron gun. *Nucl Instrum Methods Phys Res, Sect A* 2023;1051:168251.

[13] Alabdullah AI. Simulation study for the characterization of anode shape effect on the electron beam emittance of thermionic electron gun. *Optik* 2022;268:169761.

[14] Frost R, Purl O, Johnson H. Electron guns for forming solid beams of high perviance and high convergence. *Proc IRE* 1962;50(8):1800–7.

- [15] Sharma R, Sinha A, Joshi S. An improved method for the synthesis of anode aperture for Pierce guns. *IEEE Trans Electron Devices* 2001;48(2):395–7.
- [16] Pierce, J.R., *Theory and design of electron beams*. (No Title), 1954.
- [17] Rodney J, Vaughan M. Synthesis of the Pierce gun. *IEEE Trans Electron Devices* 1981;28:37–41.
- [18] Tiwary U, Basu B. Noniterative method for the synthesis of convergent pierce electron guns. *IEEE Trans Electron Devices* 1987;34(5):1218–22.
- [19] Yang C, Jia B, Zhu Z. Improved noniterative method for the synthesis of convergent pierce electron guns. *IEEE Trans Electron Devices* 2006;53(11):2849–52.
- [20] Zhang L, Cross AW. Potentials of machine learning in vacuum electronic devices demonstrated by the design of a magnetron injection gun. *IEEE Trans Electron Devices* 2021;68(6):3028–33.
- [21] Ou Y, et al. Optimum design of electron gun for 0.22-THz traveling wave tubes. *IEEE Trans Microw Theory Tech* 2021;70(1):307–14.
- [22] Bahuguna S, Mukhopadhyay S, Tiwari A. Compressed Sensing Artificial Neural Network for reactor core flux mapping. *IEEE Trans Nucl Sci* 2018;65(8):2240–9.
- [23] Chen Y, et al. A survey of accelerator architectures for deep neural networks. *Engineering* 2020;6(3):264–74.
- [24] Garnett, R., et al., *Application of Artificial Intelligence and Machine Learning to Accelerators*. 2023: Frontiers Media SA.
- [25] Blower S, et al. Evaluating and mitigating neutrons effects on COTS EdgeAI accelerators. *IEEE Trans Nucl Sci* 2021;68(8):1719–26.
- [26] Keller PE, Kouzes RT. Gamma Spectral Analysis via Neural Networks. *IEEE*; 1994.
- [27] Taylor, J.G., *Neural networks and their applications*. 1996.
- [28] Gallant AR, White H. On learning the derivatives of an unknown mapping with multilayer feedforward networks. *Neural Netw* 1992;5(1):129–38.
- [29] Hanus R, Zych M, Jaszczur M. Computational intelligence approach for liquid-gas flow regime classification based on frequency domain analysis of signals from scintillation detectors. *International Work-Conference on Artificial Neural Networks*. Springer; 2019.
- [30] Raju VG, et al. Study the Influence of Normalization/transformation Process on the Accuracy of Supervised Classification. *IEEE*; 2020.
- [31] Ozsahin DU, et al. Impact of Feature Scaling on Machine Learning Models for the Diagnosis of Diabetes. *IEEE*; 2022.
- [32] Elfaki EA, Ahmed AH. Prediction of electrical output power of combined cycle power plant using regression ANN model. *Journal of Power and Energy Engineering* 2018;6(12):17.
- [33] Yen M-H, et al. Application of the deep learning for the prediction of rainfall in Southern Taiwan. *Sci Rep* 2019;9(1):12774.
- [34] Gavin, H.P., *The Levenberg-Marquardt algorithm for nonlinear least squares curve-fitting problems*. Department of Civil and Environmental Engineering Duke University August, 2019. 3.
- [35] Azar AT. Fast neural network learning algorithms for medical applications. *Neural Comput & Applic* 2013;23(3):1019–34.
- [36] Lee Rodgers J, Nicewander WA. Thirteen ways to look at the correlation coefficient. *Am Stat* 1988;42(1):59–66.
- [37] Zhou H, et al. A new sampling method in particle filter based on Pearson correlation coefficient. *Neurocomputing* 2016;216:208–15.